Introduction to the Laplace Transform

Qi Xuan
Zhejiang University of Technology
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Structure

- Definition of the Laplace Transform
- The Step Function
- The Impulse Function
- Functional Transforms
- Operational Transforms
- Applying the Laplace Transform
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Why do we need another analytical technique?

- We wish to consider the transient behavior of circuits whose describing equations consist of *more than a single node-voltage or mesh-current differential equation*.
- We wish to determine the transient response of circuits whose *signal sources vary in ways more complicated than the simple dc level jumps*.
- We can use the Laplace transform to introduce the concept of the transfer function as a tool for *analyzing the steady-state sinusoidal response of a circuit when the frequency of the sinusoidal source is varied*.
- we wish to *relate, in a systematic fashion, the time-domain behavior of a circuit to its frequency-domain behavior*.

Definition of the Laplace Transform

Laplace Transform:

$$F(s) = \mathcal{L}\{f(t)\} = \int_0^\infty f(t)e^{-st}\,dt$$

Frequency domain

Time domain

The Laplace transform transforms the problem from the time domain to the frequency domain. *After obtaining the frequency domain expression for the unknown, we inverse-transform it back to the time domain.*

In circuit analysis, we use the Laplace transform to transform a set of *integrodifferential equations* from the time domain to a set of *algebraic equations* in the frequency domain.

General Comments

a. The upper limit of the integral is infinite. Thus, we are confronted immediately with the question of whether the integral converges. In other words, does a given $f(t)$ have a Laplace transform? *We are just interested in the functions that have Laplace transforms.*

b. Because the lower limit on the integral is zero, *the Laplace transform ignores $f(t)$ for negative values of t, which is referred as* ***one-sided****, or* ***unilateral****,* ***Laplace transform*****.** *Moreover, we integrate the function from 0^-.*

A **functional transform:** the Laplace transform of a specific function, such as $\sin \omega t$, t, e^{-at}, and so on.

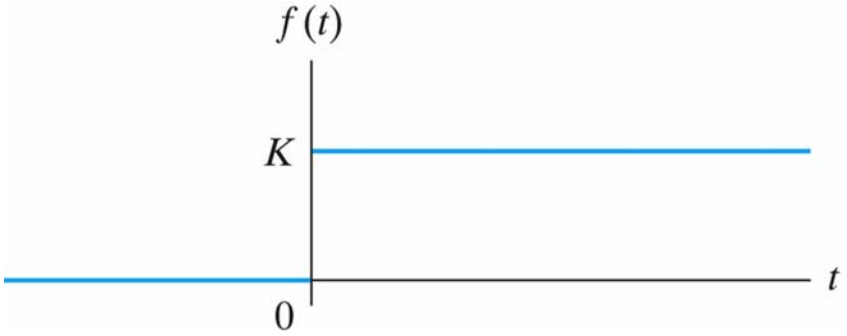
An **operational transform:** a general mathematical property of the Laplace transform, e.g., finding the transform of the derivative of $f(t)$.

The Step Function

Step function:

$$Ku(t) = 0, \quad t < 0,$$

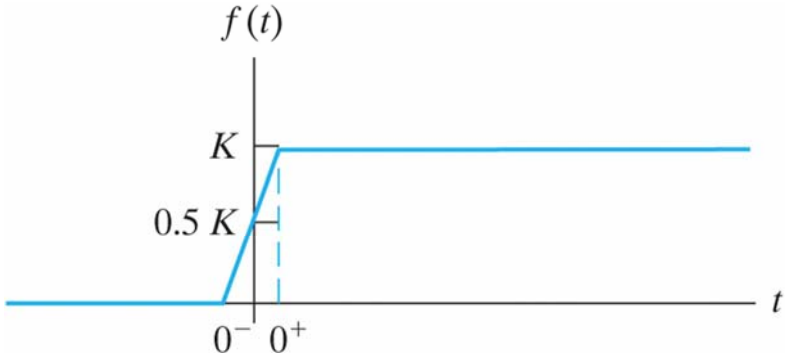
$$Ku(t) = K, \quad t > 0.$$



If K is 1, the function is the **unit step.**

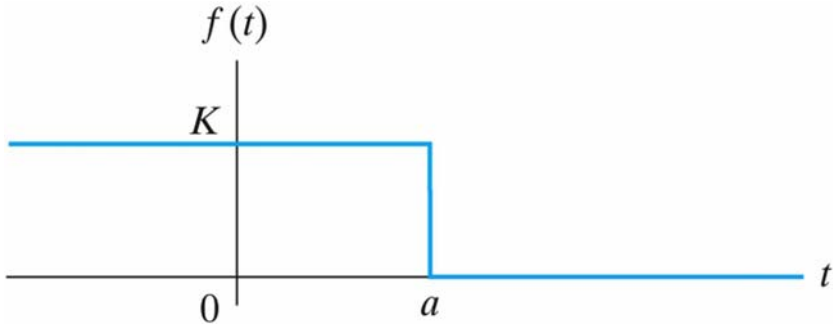
The step function is not defined *at* $t = 0$. In situations where we need to define the transition between 0^- and 0^+, we assume that it is linear and satisfies

$$Ku(0) = 0.5K$$



A discontinuity may occur at some time other than $t = 0$; for example, in sequential switching. A step that occurs *at* $t = a$ is expressed as $Ku(t - a)$.Thus

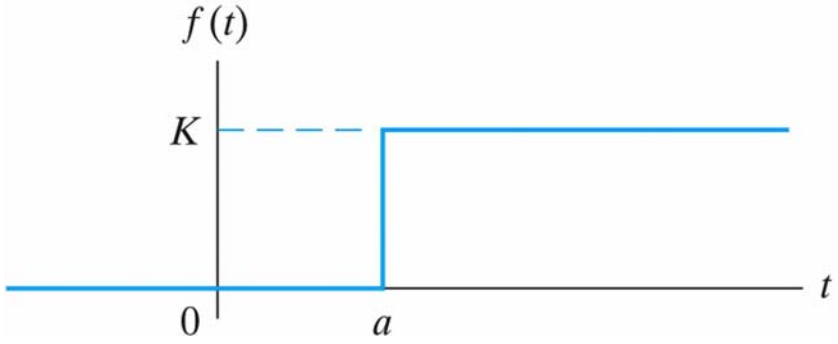
$$Ku(t - a) = 0, \quad t < a,$$
$$Ku(t - a) = K, \quad t > a.$$



If $a > 0$, the step occurs to the right of the origin, and if $a < 0$, the step occurs to the left of the origin.

A step function equal to K for $t < a$ is written as $Ku(a - t)$. Thus

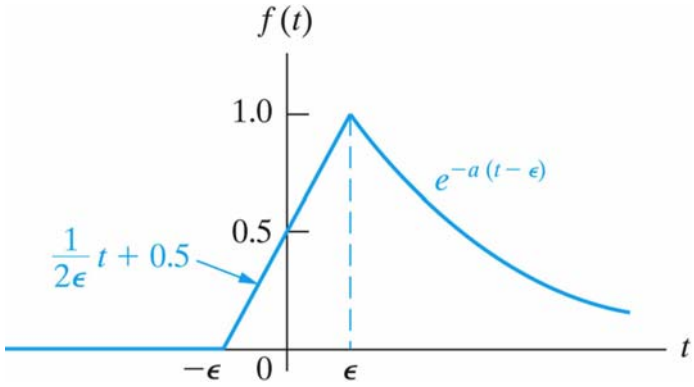
$$Ku(a - t) = K, \quad t < a,$$
$$Ku(a - t) = 0, \quad t > a.$$



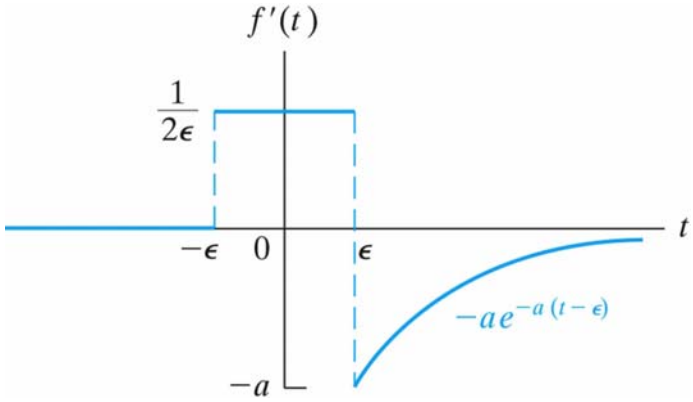
One example useful in circuit analysis is a ***finite-width pulse****, which we can create by adding two step functions.*

The Impulse Function

An **impulse** is a signal of infinite amplitude and zero duration. *Such signals don't exist in nature, but some circuit signals come very close to approximating this definition, so we find a mathematical model of an impulse useful.* Impulse function enables us to define the derivative at a discontinuity, and thus to define the Laplace transform of that derivative.



1. The amplitude approaches infinity.
2. The duration of the function approaches zero.
3. The area under the variable-parameter function is constant as the parameter changes.



Between $\pm\varepsilon$, $f'(t) \rightarrow 0$, the duration $2\varepsilon \rightarrow 0$, and the area under $f'(t) \rightarrow 1$, as $\varepsilon \rightarrow 0$, thus $f'(t)$ approaches to a unit impulse function:

$$f'(0) \rightarrow \delta(t) \quad \text{as } \epsilon \rightarrow 0$$

Sifting Property

Mathematically, the **impulse function** is defined as

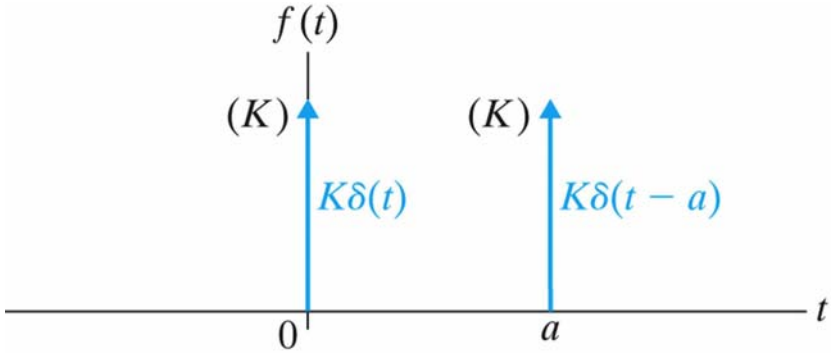
$$\int_{-\infty}^{\infty} K\delta(t)dt = K$$

$$\delta(t) = 0, \quad t \neq 0$$

The area under the impulse function is constant, and this area represents the strength of the impulse.

The **sifting property** is expressed as

$$\int_{-\infty}^{\infty} f(t)\delta(t - a)\,dt = f(a)$$



where the function f(t) is assumed to be continuous at t = a; that is, at the location of the impulse. (The validation is shown from Eq. 12.12 to Eq. 12.13)

Laplace Transform

$e^{-st} \rightarrow 1$, as $t \rightarrow 0$

$$\mathcal{L}\{\delta(t)\} = \int_{0^-}^{\infty} \delta(t)e^{-st}\,dt = \int_{0^-}^{\infty} \delta(t)\,dt = 1$$

$$L\{\delta'(t)\} = \int_{0^-}^{\infty} \delta'(t)e^{-st}dt = \int_{0^-}^{\infty} e^{-st}d\delta(t)$$

$$= e^{-st}\delta(t)|_{0^-}^{\infty} - \int_{0^-}^{\infty} \delta(t)de^{-st} = -\int_{0^-}^{\infty} -s\delta(t)e^{-st}dt = sL\{\delta(t)\} = s$$

$$L\left(\delta^n(t)\right) = sL\left(\delta^{n-1}(t)\right) = s^2L\left(\delta^{n-2}(t)\right) = \cdots = s^nL\left(\delta(t)\right) = s^n$$

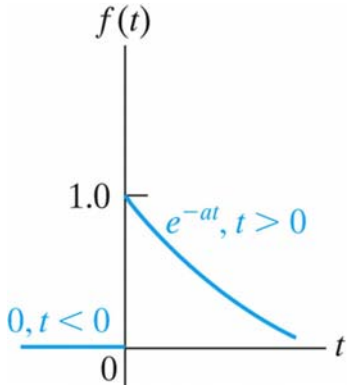
$$\delta(t) = \frac{du(t)}{dt}$$

Functional Transforms

$$\mathcal{L}\{\delta(t)\} = \int_{0^-}^{\infty} \delta(t)e^{-st}\,dt = \int_{0^-}^{\infty} \delta(t)dt = 1$$

$$\mathcal{L}\{u(t)\} = \int_{0^-}^{\infty} f(t)e^{-st}\,dt = \int_{0^-}^{\infty} 1e^{-st}dt = \frac{e^{-st}}{-s}\bigg|_{0^-}^{\infty} = \frac{1}{s}$$

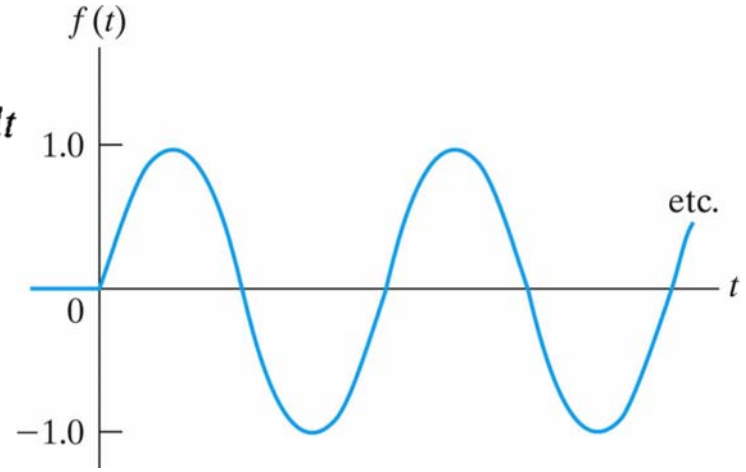
$$\mathcal{L}\{e^{-at}\} = \int_{0^+}^{\infty} e^{-at}e^{-st}\,dt = \int_{0^+}^{\infty} e^{-(a+s)t}\,dt = \frac{1}{s+a}$$



$$\mathcal{L}\{\sin \omega t\} = \int_{0^-}^{\infty} (\sin \omega t)e^{-st}\,dt = \int_{0^-}^{\infty} \left(\frac{e^{j\omega t} - e^{-j\omega t}}{2j}\right)e^{-st}\,dt$$

$$= \int_{0^-}^{\infty} \frac{e^{-(s-j\omega)t} - e^{-(s+j\omega)t}}{2j}\,dt$$

$$= \frac{1}{2j}\left(\frac{1}{s-j\omega} - \frac{1}{s+j\omega}\right) = \frac{\omega}{s^2+\omega^2}$$



Type	$f(t)$ $(t > 0-)$	$F(s)$
(impulse)	$\delta(t)$	1
(step)	$u(t)$	$\frac{1}{s}$
(ramp)	t	$\frac{1}{s^2}$
(exponential)	e^{-at}	$\frac{1}{s + a}$
(sine)	$\sin \omega t$	$\frac{\omega}{s^2 + \omega^2}$
(cosine)	$\cos \omega t$	$\frac{s}{s^2 + \omega^2}$
(damped ramp)	te^{-at}	$\frac{1}{(s + a)^2}$
(damped sine)	$e^{-at} \sin \omega t$	$\frac{\omega}{(s + a)^2 + \omega^2}$
(damped cosine)	$e^{-at} \cos \omega t$	$\frac{s + a}{(s + a)^2 + \omega^2}$

Operational Transforms

Multiplication by a Constant

$$\mathcal{L}\{f(t)\} = F(s) \quad \Rightarrow \quad \mathcal{L}\{Kf(t)\} = KF(s)$$

Addition (Subtraction)

$$\mathcal{L}\{f_1(t)\} = F_1(s)$$
$$\mathcal{L}\{f_2(t)\} = F_2(s) \quad \Rightarrow \quad \mathcal{L}\{f_1(t) + f_2(t) - f_3(t)\} = F_1(s) + F_2(s) - F_3(s)$$
$$\mathcal{L}\{f_3(t)\} = F_3(s)$$

Differentiation

$$\mathcal{L}\left\{\frac{df(t)}{dt}\right\} = \int_{0^-}^{\infty}\left[\frac{df(t)}{dt}\right]e^{-st}\,dt = e^{-st}f(t)\Big|_{0^-}^{\infty} - \int_{0^-}^{\infty} f(t)(-se^{-st}dt)$$

$$= -f(0^-) + s\int_{0^-}^{\infty} f(t)e^{-st}dt = sF(s) - f(0^-)$$

$$\mathcal{L}\left\{\frac{d^nf(t)}{dt^n}\right\} = s^nF(s) - s^{n-1}f(0^-) - s^{n-2}\frac{df(0^-)}{dt} - s^{n-3}\frac{d^2f(0^-)}{dt^2} - \cdots - \frac{d^{n-1}f(0^-)}{dt^{n-1}}$$

Integration

$$\mathcal{L}\left\{\int_{0^-}^{t} f(x)\,dx\right\} = \int_{0^-}^{\infty}\left[\int_{0^-}^{t} f(x)\,dx\right]e^{-st}\,dt$$

$$u = \int_{0^-}^{t} f(x)\,dx \quad \Rightarrow \quad du = f(t)\,dt$$

$$v = -\frac{e^{-st}}{s} \quad \Rightarrow \quad dv = e^{-st}\,dt$$

$$\mathcal{L}\left\{\int_{0^-}^{t} f(x)\,dx\right\} = \underbrace{-\frac{e^{-st}}{s}\int_{0^-}^{t} f(x)\,dx\Bigg|_{0^-}^{\infty}}_{=0} + \int_{0^-}^{\infty}\frac{e^{-st}}{s} f(t)\,dt = \frac{F(s)}{s}$$

The evaluation at the lower limit obviously is zero, whereas *the evaluation at the upper limit is zero because we are assuming that f(t) has a Laplace transform*.

Translation in the Time Domain

$$\mathcal{L}\{f(t-a)u(t-a)\} = \int_{0^-}^{\infty} u(t-a)f(t-a)e^{-st}\,dt$$

$$= \int_{a}^{\infty} f(t-a)e^{-st}\,dt \qquad u(t-a) = 1 \text{ for } t > a$$

let $x = t - a$

$x = 0$ when $t = a$

$x = \infty$ when $t = \infty$

$dx = dt$

$$\mathcal{L}\{f(t-a)u(t-a)\} = \int_{0}^{\infty} f(x)e^{-s(x+a)}\,dx$$

$$= e^{-sa}\int_{0}^{\infty} f(x)e^{-sx}\,dx = e^{-as}F(s)$$

Translation in the Frequency Domain

$$\mathcal{L}\{e^{-at} f(t)\} = F(s + a)$$

(Problem 12.14)

$$\mathcal{L}\{\cos \omega t\} = \frac{s}{s^2 + \omega^2}$$

$$\mathcal{L}\{e^{-at} \cos \omega t\} = \frac{s + a}{(s + a)^2 + \omega^2}$$

Scaling Changing

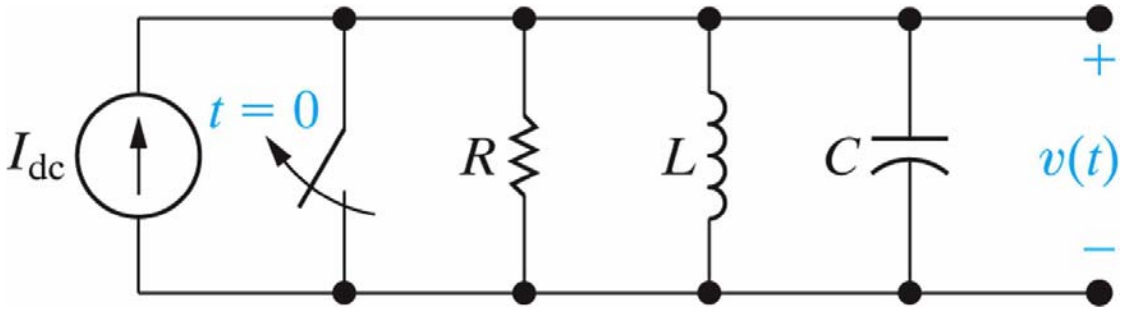
$$\mathcal{L}\{f(at)\} = \frac{1}{a} F\left(\frac{s}{a}\right), \quad a > 0$$

(Problem 12.16)

$$\mathcal{L}\{\cos t\} = \frac{s}{s^2 + 1}$$

$$\mathcal{L}\{\cos \omega t\} = \frac{1}{\omega} \frac{s/\omega}{(s/\omega)^2 + 1} = \frac{s}{s^2 + \omega^2}$$

Applying the Laplace Transform



$$\frac{v(t)}{R} + \frac{1}{L}\int_0^t v(x)\,dx + C\frac{dv(t)}{dt} = I_{dc}u(t)$$

$$\frac{V(s)}{R} + \frac{1}{L}\frac{V(s)}{s} + C[sV(s) - v(0^-)] = I_{dc}\left(\frac{1}{s}\right)$$

$$V(s)\left(\frac{1}{R} + \frac{1}{sL} + sC\right) = \frac{I_{dc}}{s} \quad \Rightarrow \quad V(s) = \frac{I_{dc}/C}{s^2 + (1/RC)s + (1/LC)}$$

$$v(t) = \mathcal{L}^{-1}\{V(s)\}$$

Inverse transform

Inverse Transforms

- A *rational function* of *s*: the function that can be expressed in the form of a ratio of two polynomials in *s* such that no nonintegral powers of *s* appear in the polynomials.

$$F(s) = \frac{N(s)}{D(s)} = \frac{a_n s^n + a_{n-1} s^{n-1} + \cdots + a_1 s + a_0}{b_m s^m + b_{m-1} s^{m-1} + \cdots + b_1 s + b_0}$$

- *Proper rational function*: $m > n$
- *Improper rational function*: $m \leq n$

Partial Fraction Expansion

A proper rational function is expanded into a sum of partial fractions *by writing a term or a series of terms for each root of* $D(s)$. *Thus* $D(s)$ *must be in factored form* before we can make a partial fraction expansion.

$$\frac{s+6}{s(s+3)(s+1)^2} = \frac{K_1}{s} + \frac{K_2}{s+3} + \frac{K_3}{(s+1)^2} + \frac{K_4}{s+1}$$

multiple root

The denominator $D(s)$ has four roots, two distinct roots at $s = 0$ and $s = -3$, and a multiple root $s = -1$, then we have

$$\mathcal{L}^{-1}\left\{\frac{s+6}{s(s+3)(s+1)^2}\right\} = (K_1 + K_2 e^{-3t} + K_3 t e^{-t} + K_4 e^{-t})u(t)$$

- We used the *identity sign* to emphasize that expanding a rational function into a sum of partial fractions establishes an identical equation. Thus *both sides of the equation must be the same for all values of the variable* s. Also, *the identity relationship must hold when both sides are subjected to the same mathematical operation*.
- All that remains is to establish a technique for *determining the coefficients* (K_1, K_2, K_3, …) generated by making a partial fraction expansion. There are four general forms this problem can take. Specifically, the roots of $D(s)$ are either *(1) real and distinct; (2) complex and distinct; (3) real and repeated; or (4) complex and repeated.*
- Be sure to verify that *the rational function is proper*.

Distinct Real Roots of $D(s)$

$$F(s) = \frac{96(s+5)(s+12)}{s(s+8)(s+6)} \equiv \frac{K_1}{s} + \frac{K_2}{s+8} + \frac{K_3}{s+6}$$

To find the value of K_1 we multiply both sides by s and then evaluate both sides at $s = 0$

$$\left.\frac{96(s+5)(s+12)}{(s+8)(s+6)}\right|_{s=0} \equiv K_1 + \left.\frac{K_2 s}{s+8}\right|_{s=0} + \left.\frac{K_3 s}{s+6}\right|_{s=0} \Rightarrow K_1 = 120$$

Similarly, we can get $K_2 = -72$, $K_3 = 48$, then

$$\frac{96(s+5)(s+12)}{s(s+8)(s+6)} \equiv \frac{120}{s} + \frac{48}{s+6} - \frac{72}{s+8}$$

Check!

$$\Rightarrow \mathcal{L}^{-1}\left\{\frac{96(s+5)(s+12)}{s(s+8)(s+6)}\right\} = (120 + 48e^{-6t} - 72e^{-8t})u(t)$$

Distinct Complex Roots of $D(S)$

$$\frac{100(s+3)}{(s+6)(s^2+6s+25)} \equiv \frac{K_1}{s+6} + \frac{K_2}{s+3-j4} + \frac{K_3}{s+3+j4}$$

$$K_1 = \left.\frac{100(s+3)}{s^2+6s+25}\right|_{s=-6} = \frac{100(-3)}{25} = -12$$

$$K_2 = \left.\frac{100(s+3)}{(s+6)(s+3+j4)}\right|_{s=-3+j4} = \frac{100(j4)}{(3+j4)(j8)} = 6 - j8 = 10e^{-j53.13^\circ}$$

$$K_3 = \left.\frac{100(s+3)}{(s+6)(s+3-j4)}\right|_{s=-3-j4} = \frac{100(-j4)}{(3-j4)(-j8)} = 6 + j8 = 10e^{j53.13^\circ}$$

$$\frac{100(s+3)}{(s+6)(s^2+6s+25)} = \frac{-12}{s+6} + \frac{10\angle -53.13^\circ}{s+3-j4} + \frac{10\angle 53.13^\circ}{s+3+j4}$$

$$\mathcal{L}^{-1}\left\{\frac{100(s+3)}{(s+6)(s^2+6s+25)}\right\} = (-12e^{-6t} + 10e^{-j53.13^\circ}e^{-(3-j4)t} + 10e^{j53.13^\circ}e^{-(3+j4)t})u(t)$$

Can be further simplified

Whenever $D(s)$ contains distinct complex roots—that is, factors of the form $(s + \alpha - j\beta)(s + \alpha + j\beta)$—a pair of terms have the following form

$$\frac{K}{s + \alpha - j\beta} + \frac{K^*}{s + \alpha + j\beta}$$

$$K = |K|e^{j\theta} = |K|\angle\theta^\circ$$

$$K^* = |K|e^{-j\theta} = |K|\angle -\theta^\circ$$

$$\mathcal{L}^{-1}\left\{\frac{K}{s + \alpha - j\beta} + \frac{K^*}{s + \alpha + j\beta}\right\} = 2|K|e^{-\alpha t}\cos(\beta t + \theta)$$

$$\mathcal{L}^{-1}\left\{\frac{100(s + 3)}{(s + 6)(s^2 + 6s + 25)}\right\} = [-12e^{-6t} + 20e^{-3t}\cos(4t - 53.13^\circ)]u(t)$$

Repeated Real Roots of $D(s)$

$$\frac{100(s+25)}{s(s+5)^3} = \frac{K_1}{s} + \frac{K_2}{(s+5)^3} + \frac{K_3}{(s+5)^2} + \frac{K_4}{s+5}$$

To find K_2, we multiply both sides by $(s+5)^3$ and then evaluate both sides at -5 :

$$\left.\frac{100(s+25)}{s}\right|_{s=-5} = \left.\frac{K_1(s+5)^3}{s}\right|_{s=-5} + K_2 + K_3(s+5)\big|_{s=-5} + \left.K_4(s+5)^2\right|_{s=-5}$$

$$K_2 = -400$$

To find K_3, we first must multiply both sides by $(s+5)^3$. Next we differentiate both sides once with respect to s and then evaluate at $s = -5$:

$$K_3 = -100$$

$$\frac{d}{ds}\left[\frac{100(s+25)}{s}\right]_{s=-5} = \frac{d}{ds}\left[\frac{K_1(s+5)^3}{s}\right]_{s=-5} + \frac{d}{ds}[K_2]_{s=-5} + \frac{d}{ds}[K_3(s+5)]_{s=-5} + \frac{d}{ds}[K_4(s+5)^2]_{s=-5}$$

To find K_4 we first multiply both sides by $(s+5)^3$. Next we differentiate both sides *twice* with respect to s and then evaluate both sides at $s = -5$.

$$K_4 = -20$$

$$100\frac{d}{ds}\left[-\frac{25}{s^2}\right]_{s=-5} = K_1\frac{d}{ds}\left[\frac{(s+5)^2(2s-5)}{s^2}\right]_{s=-5} + 0 + \frac{d}{ds}[K_3]_{s=-5} + \frac{d}{ds}[2K_4(s+5)]_{s=-5}$$

$$\frac{100(s+25)}{s(s+5)^3} = \frac{20}{s} - \frac{400}{(s+5)^3} - \frac{100}{(s+5)^2} - \frac{20}{s+5}$$

$$\mathcal{L}^{-1}\left\{\frac{100(s+25)}{s(s+5)^3}\right\} = [20 - 200t^2e^{-5t} - 100te^{-5t} - 20e^{-5t}]u(t)$$

Repeated Complex Roots of $D(s)$

We handle repeated complex roots in the same way that we did repeated real roots; *the only difference is that the algebra involves complex numbers*. Recall that *complex roots always appear in conjugate pairs and that the coefficients associated with a conjugate pair are also conjugates*, so that only half the Ks need to be evaluated.

$$F(s) = \frac{768}{(s^2 + 6s + 25)^2}$$

$$F(s) = \frac{768}{(s + 3 - j4)^2(s + 3 + j4)^2}$$

$$= \frac{K_1}{(s + 3 - j4)^2} + \frac{K_2}{s + 3 - j4} + \frac{K_1^*}{(s + 3 + j4)^2} + \frac{K_2^*}{s + 3 + j4}$$

$$K_1 = \left.\frac{768}{(s + 3 + j4)^2}\right|_{s=-3+j4} = \frac{768}{(j8)^2} = -12 \qquad K_1^* = -12$$

$$K_2 = \frac{d}{ds}\left[\frac{768}{(s + 3 + j4)^2}\right]_{s=-3+j4} = -j3 = 3\angle -90^\circ \qquad K_2^* = j3 = 3\angle 90^\circ$$

$$F(s) = \left[\frac{-12}{(s+3-j4)^2} + \frac{-12}{(s+3+j4)^2}\right] + \left(\frac{3\angle -90^\circ}{s+3-j4} + \frac{3\angle 90^\circ}{s+3+j4}\right)$$

$$f(t) = [-24te^{-3t}\cos 4t + 6e^{-3t}\cos(4t - 90^\circ)]u(t)$$

Multiple real roots

$$\mathcal{L}^{-1}\left\{\frac{K}{(s+a)^r}\right\} = \frac{Kt^{r-1}e^{-at}}{(r-1)!}u(t)$$

Multiple complex roots

$$\mathcal{L}^{-1}\left\{\frac{K}{(s+\alpha-j\beta)^r} + \frac{K^*}{(s+\alpha+j\beta)^r}\right\} = \left[\frac{2|K|t^{r-1}}{(r-1)!}e^{-\alpha t}\cos(\beta t + \theta)\right]u(t)$$

Improper Rational Functions

$$F(s) = \frac{s^4 + 13s^3 + 66s^2 + 200s + 300}{s^2 + 9s + 20}$$

$$F(s) = s^2 + 4s + 10 + \frac{30s + 100}{s^2 + 9s + 20}$$

$$\frac{30s + 100}{s^2 + 9s + 20} = \frac{30s + 100}{(s + 4)(s + 5)} = \frac{-20}{s + 4} + \frac{50}{s + 5}$$

$$F(s) = s^2 + 4s + 10 - \frac{20}{s + 4} + \frac{50}{s + 5}$$

$$f(t) = \frac{d^2\delta(t)}{dt^2} + 4\frac{d\delta(t)}{dt} + 10\delta(t) - (20e^{-4t} - 50e^{-5t})u(t)$$

Summary

- Laplace transform
- Step function (Unit step function)
- Impulse function (Unit impulse function)
- Functional transform
- Operational transform